[11] and Pathways Language Model (PaLM) [20] contains more than 100 billion parameters, which require substantial computational power to train. Fine-tuning or re-training these models necessitates access to high-performance GPUs or specialized hardware, such as TPUs, which can be expensive and difficult to obtain, especially for individual researchers or smaller organizations.

Over the past few years, significant research has been conducted on domain specialization techniques for LLMs. Many methods focus on generic technical contributions, adaptable to specific domains with minor modifications and access to domain-specific information. However, cross-referencing these techniques across different application domains remains a challenge, as does the absence of a systematic standardization and summary of methods for evaluating various domain specialization techniques. This lack of clarity creates obstacles for non-AI professionals and obfuscates existing bottlenecks, pitfalls, open problems, and potential future research directions. To surmount these obstacles and harness artificial intelligence for more effectively accomplishing tasks across various domains, this survey paper offers a comprehensive and systematic review of the current state-of-the-art LLM domain specialization. The major contributions of this paper include:

- **A systematic categorization and taxonomy of LLMs domain specialization techniques**: We comprehensively classify existing methods based on different levels (i.e., black-box, grey-box, and white-box) of accessibility to the LLM and organize their corresponding techniques into a taxonomy. We discuss details, relationships, pros, and cons among different subcategories. The proposed taxonomy is designed to assist domain experts in identifying the most suitable techniques for their target problem settings.
- **A comprehensive categorization and summarization of major application domains**: We debut the taxonomy of representative application domains that domain-specialized LLMs can enhance. The practical significance and open challenges for each application domain or subdomain are elucidated, allowing for easy mapping to the proposed technique taxonomy. Researchers and various domain experts could cross-reference additional application domains for evaluating their newly proposed methods while expanding their advanced techniques to encompass new application domains.
- **An insightful discussion of the current status of research in this area and future trends**. Based on the comprehensive and systematic survey and investigation of existing domain specialization techniques and applications, an overall picture and trends of LLM domain specialization have been outlined and discussed. The open challenges and The paper concludes by presenting fresh insights into the bottlenecks, open problems, as well as a discussion of possible future directions.

1.1 Related Surveys

This section briefly outlines previous surveys relevant to the domain specialization of LLMs in three categories: (1) fundamental overview of PLMs and LLMs; (2). techniques on domain adaptation and generalization of PLMs; and (3). Specializing language models for specific domains.

Fundamental overview of PLMs and LLMs. While comprehensive reviews [101, 123] of PLMs and their use in diverse NLP tasks exist, they don't necessarily apply to LLMs due to differences between the two. Given the recent growth in popularity and effectiveness of LLMs, several review papers have emerged, addressing various LLM aspects. Some focus on fundamental LLM components [86, 173, 186], others on the history and potential applications of generative AI [13, 180], and a few [100] on enhancing LLMs with reasoning capabilities. However, a comprehensive review and technical taxonomy of LLM domain specialization are yet to be provided.

Domain adaptation and generalization of PLMs. Surveys [31, 42] examine how to effectively and efficiently adapt PLMs to specific domains, such as adding a layer to the model or updating the model parameters. However, most of these techniques don't apply to LLMs because of the inaccessibility of their architecture and parameter space. Also, updating knowledge in LLMs is challenging due to computational costs and the need for efficient optimization strategies.

Specializing language models for specific domains. Recent review papers have emphasized the benefits and necessity of customizing LLMs for specific domains. Risks linked with applying generic LLMs to areas like medical education have been noted in [24, 133], including lack of originality and inaccuracies. Practical considerations for legal domain-specific language models have also been suggested in [139]. In the finance sector, initial steps towards a finance-specialized LLM have shown improved performance on financial tasks without compromising general benchmarks [169]. These advances highlight the need for a comprehensive review and technical taxonomy of domain specialization techniques to assist different sectors in effectively employing LLMs for their unique tasks.

2 TAXONOMY OF DOMAIN SPECIALIZATION

Large language models are typically referred to as large-scale pre-trained language models (PLMs) based on the Transformer architecture [101, 123]. Empirical evidence suggests that scaling pre-trained language models, such as increasing the model size or data size, frequently results in enhanced model capacity for downstream tasks. In this section, we begin by reviewing the fundamental concepts of PLMs and proceed to present a comprehensive taxonomy of existing techniques aimed at specializing large language models for specific domains.

2.1 Background

Specifically, PLM is a type of neural network pre-trained on a large corpus of text data to learn linguistic patterns, structures, and semantics. The input and output of PLMs can be described as follows. In LLMs, the *input* is a text sequence that serves as context for understanding and processing. To clarify the task, a *prompt*, an additional sentence or query, is often included. These prompts, designed based on the NLP task, provide a premise or task explanation. For instance, in text summarization, a prompt like "Summarize the key points in the following passage:" could precede the input passage. The *output* is the text sequence or prediction generated in response to the input. Depending on the task, this could be an answer to a question or a sentiment label, and may require post-processing like token decoding or label extraction for final presentation. As LLMs are typically scaled-up versions of PLMs, they follow the similar architecture design of PLMs, which come in three main flavors: *encoder-only*, *encoder-decoder*, and *decoder-only* architectures. This brief introduction will provide an overview of these PLMs architectures, and discuss their differences and commonalities.

- *Encoder-only Language Models* process input text into vector representations without an explicit decoding phase to generate new text. Instead, they transform and embed text into a high-dimensional space. These models are primarily designed to capture and understand the patterns and semantics in the input data. They are extensively used for tasks such as text classification, sentiment analysis, and clustering. One of the notable examples is BERT [30], which extracts context-rich embeddings for downstream tasks by pre-training on a masked language modeling objective.
- *Encoder-Decoder Language Models* consist of an encoder that processes input text into vector representations and a decoder that generates output text from these representations. They employ cross-entropy loss as the objective function, comparing the actual and predicted target sequences. These PLMs are often used for sequence-to-sequence tasks like machine translation and summarization, with T5 [125] being a notable example.

- *Decoder-only Language Models*, like GPT [124], are autoregressive language models that generate the next word in a sequence based on previous words. They map a sequence of tokens to a vector representation and generate contextually relevant content autoregressively, calculating the probability of the next token based on the context. This autoregressive modeling approach is particularly suitable for text-generation tasks.

2.2 Taxonomy of Domain Specialization Techniques

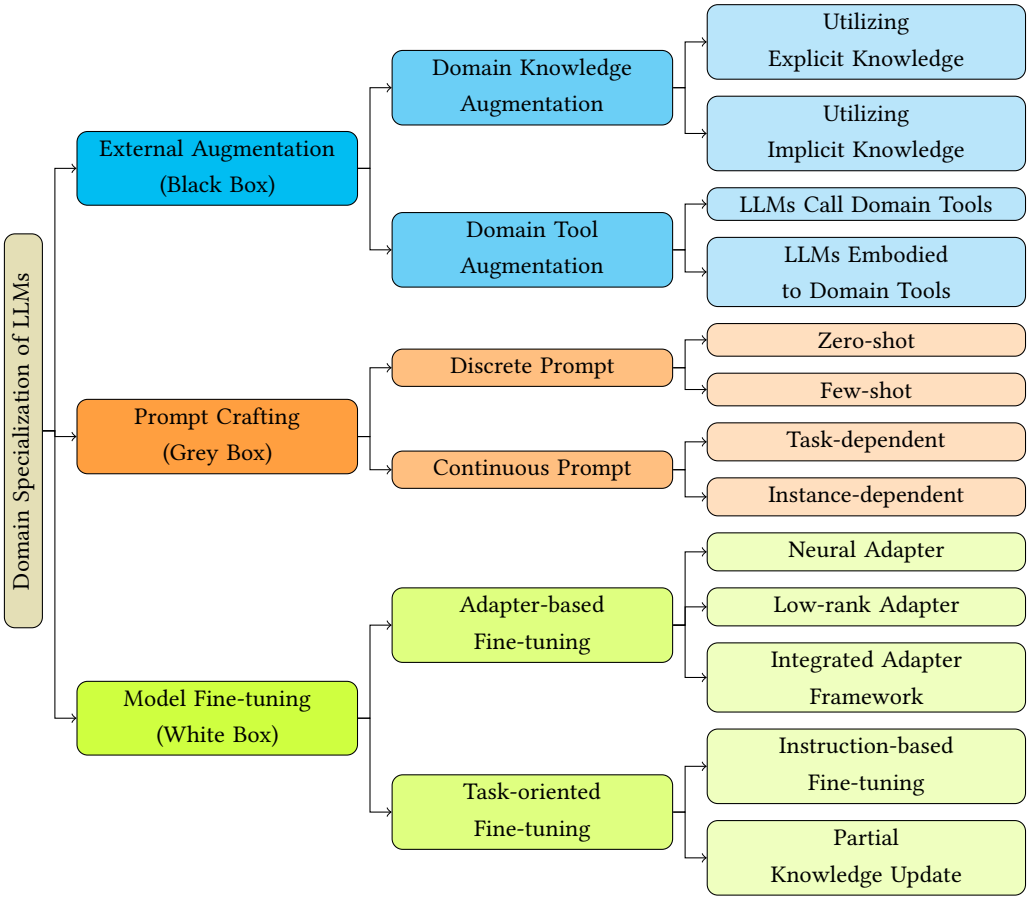

Fig. 1. The taxonomy for current techniques on LLM domain specialization.

Domain specialization of LLMs can be understood as tailoring broad, universally-trained LLMs to operate optimally within a specific field or domain. To tackle the three challenges of domain specialization mentioned in Section 1, respectively, the approaches in LLM domain specialization can be categorized into three corresponding classes of approaches: *external augmentation*, *prompt crafting*, and *model fine-tuning*. These classes correspond to *assumptions* of different levels of accessibility to LLMs, namely, no access (black box), partial access (grey box), and full access (white box). The black box *assumption* typically indicates we only have access to the model API (e.g., ChatGPT) without knowing any information but the generated output; the grey box *assumption* denotes we have limited information (e.g., the probability of generated tokens in GPT-3 API), such information can guide us to design and fine-tune a suitable